

Level 2 Charging Infrastructure for Operationally Suited Fleet Vehicles

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DSC 680 Applied Data Science

Milestone 3 – White Paper

Level 2 Charging Infrastructure for Operationally Suited Fleet Vehicles

Business Problem and Research Questions

Commercial fleet operators need an early way to estimate whether vehicles identified as operationally suitable for electrification can also be supported by Level 2 charging infrastructure. Fleet size alone can't answer that question because daily mileage, vehicle class, available charging time, charger power, and site cost can produce very different infrastructure requirements. Without a clearer estimate, a fleet could install too few chargers to support daily operations or spend more than necessary on equipment and electrical capacity.

This project builds on the operational-suitability analysis completed in an earlier fleet project. That analysis established the rules used to identify vehicle-days with operating patterns favorable for electrification. The current project reapplies those rules to create the primary group of vehicles used in the charging analysis. It then evaluates whether those vehicles can recover their estimated daily energy using different Level 2 charger powers and overnight charging windows.

Results for all quality-eligible vehicles are retained as a comparison. This makes it possible to measure how focusing on operationally suitable vehicles changes estimated energy demand and charging feasibility. Charging requirements for selected vehicles are calculated from all their usable operating days, including more demanding days, rather than only the days classified as highly suitable.

The purpose isn't to produce a construction-ready design. It's to identify deployments that appear ready for a more detailed site assessment, estimate their preliminary Level 2 charging requirements, and show which assumptions have the greatest effect on feasibility and cost. The analysis addresses five research questions:

- How does focusing on operationally suitable vehicles change deployment-level energy demand and Level 2 charging feasibility compared with including all quality-eligible vehicles?
- How much daily grid energy would the operationally suitable vehicles require at the deployment level?
- Under which combinations of Level 2 charger power and overnight charging time are the deployments screening-feasible?
- What planning-level equipment and installation costs are associated with the lowest-cost feasible configuration under the 10-hour baseline?
- How sensitive are the feasibility results to changes in the assumed energy use of the selected vehicles?

Background and History

Fleet electrification planning involves two related questions. The earlier fleet project addressed whether a vehicle's operating patterns make it a good candidate for electrification. This project addresses the next question: whether the depot's charging infrastructure can provide enough energy while the selected vehicles are parked. Prior research and federal guidance show that return-to-base fleets can be strong candidates for electrification because vehicles often follow repeatable routes and spend extended periods at a common location, but charging needs still vary with mileage, vehicle class, dwell time, and available power (Borlaug et al., 2020; U.S. Department of Energy, n.d.-b). Level 2 equipment is relevant to this screening because it can serve vehicles that return to a depot for long overnight windows without immediately assuming higher-powered direct-current fast charging.

The project uses the National Renewable Energy Laboratory's Fleet DNA composite data, which contain anonymized operating records for medium- and heavy-duty fleet vehicles (Duran et al., 2014; Lustbader et al., 2016). Fleet DNA provides real-world use patterns, but it doesn't contain complete depot arrival and departure schedules, measured electric energy use for every vehicle, charger outcomes, or site construction costs. The analysis will combine observed operating data with clearly labeled planning assumptions. The model can help identify which fleet deployments should move forward to a detailed site assessment first, but it can't replace utility review, electrical engineering, or a vehicle-by-vehicle charging schedule.

Data Explanation

The downloaded composite file contained 4,705 vehicle-day records representing 486 vehicles and 57 deployments. The input review found no duplicate vehicle-day records and no missing or nonpositive values in the fields used for the suitability screen. Vehicles with fewer than three recorded days were removed because a one- or two-day sample could overstate a vehicle's normal operating pattern. That quality rule excluded 53 vehicles and retained 433 for evaluation. The notebook performs each cleaning, transformation, and aggregation step so the reported results can be reproduced.

The earlier project evaluated operational suitability at the vehicle-day level using five rules: 20 to 120 daily miles; no more than eight driving hours; an average speed no higher than 35 miles per hour; either at least 30% zero-speed time or at least one stop per mile; and at least four trips and 30 stops. A day was labeled High Suitability when it met at least four of these five rules. This project reapplies those established rules so both analyses use the same definition of operational suitability. Because the earlier project classified individual vehicle-days rather than entire vehicles, the current analysis combines the daily results at the vehicle level. A vehicle

enters the primary charging cohort when more than half of its recorded days meet the High Suitability criteria. This 50% threshold was selected as a measure of consistency for the infrastructure analysis; it wasn't learned from the data or taken from an industry standard.

Across all usable records, 3,481 vehicle-days, or 74.0%, met the daily high-suitability definition. The majority rule selected 373 vehicles from 53 deployments. Sixty quality-eligible vehicles remained outside the primary cohort, and four deployments had no selected vehicles. Once a vehicle was selected, its charging requirement was calculated from all usable days rather than only the favorable days. This avoids lowering energy demand by discarding harder operating days after selection. Appendix A lists the main fields, transformations, and decision rules.

Methods

Operational suitability isn't remodeled in this project. The earlier project's rules are reapplied as a cohort-selection step, while the new analysis begins with estimating energy demand and testing Level 2 charging scenarios. A deterministic scenario model fits this problem because the available data support transparent calculations but don't include a historical target such as observed charger count, site cost, or project success. A predictive model would imply that those outcomes had been learned from the data when they hadn't. The scenario approach instead makes each assumption visible, allows sensitivity testing, and produces a screening result that a fleet manager can audit.

For each selected vehicle, the model uses its 90th-percentile daily mileage as the planning value. The 90th percentile was chosen to cover most observed days without sizing every deployment to a single extreme maximum. Mileage is converted to battery energy with class-level factors ranging from 0.55 kilowatt-hours per mile for Class 2 to 2.10 for Class 8, then

divided by 90% charging efficiency to estimate grid energy. These factors are AFLEET-informed planning assumptions rather than measurements from the Fleet DNA records (Argonne National Laboratory, n.d.). They provide a consistent first-pass conversion, but actual energy use will change with vehicle model, payload, weather, terrain, auxiliary loads, and driver behavior.

The model tests 7.2, 11.5, and 19.2 kilowatt Level 2 chargers because they represent low, middle, and upper-power options within the Level 2 range and make the tradeoff between charging time and infrastructure visible. It tests 8-, 10-, and 12-hour overnight windows because Fleet DNA doesn't provide complete depot dwell schedules. The 10-hour window is the baseline because it is the midpoint of the tested range, not because the data showed that 10 hours was the most common dwell time. Only 85% of each window is treated as usable to allow for late returns, staging, plug-in delays, and other interruptions.

A deployment is labeled screening-feasible only when three conditions are met: the charger configuration can deliver the deployment's aggregate planning energy, the highest-demand vehicle can receive its own required energy within the window, and the required port count doesn't exceed the number of selected vehicles. The model then chooses the lowest-cost feasible option for the 10-hour baseline, breaking ties in favor of less connected power. This doesn't prove that every vehicle can be assigned to a port at the correct time. It also doesn't confirm transformer capacity, electrical service, parking layout, or code compliance. Those checks belong in a later site-specific design.

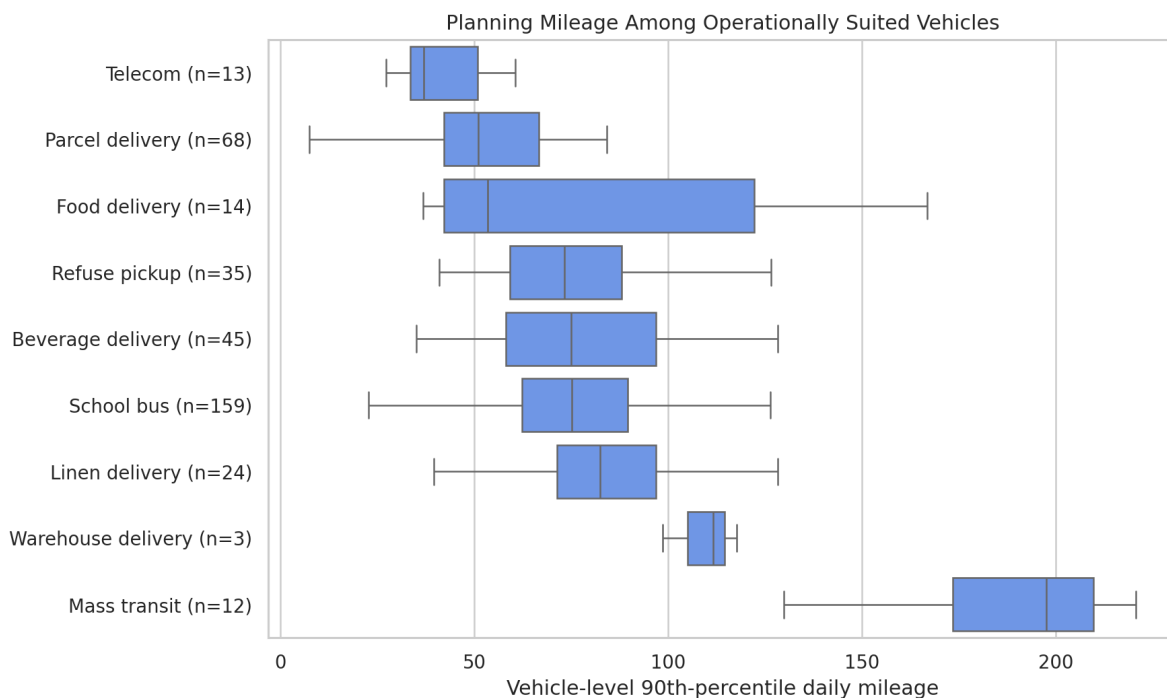
Planning costs combine charger equipment with low, typical, and high per-port installation assumptions. The typical case uses \$2,500 per port, while the low and high cases use \$600 and \$12,700. Equipment costs range from \$2,000 to \$5,000 per port based on charger power. These values create a comparable screening range; they aren't vendor quotes and don't

include major utility upgrades, transformer work, unusual trenching, permits, networking, or site-specific labor. Monte Carlo simulation was considered as a way to represent uncertainty, but the current inputs don't have defensible probability distributions. The low, typical, high, and $\pm 15\%$ sensitivity cases are easier to explain and audit at this stage.

Analysis

The selected vehicles don't form a single operating profile. Figure 1 shows meaningful differences in 90th-percentile mileage across vocations, including substantial within-vocation variation. This supports using vehicle-level planning mileage instead of assigning one value to an entire vocation. It also explains why vocation labels alone shouldn't determine which fleets advance to engineering review.

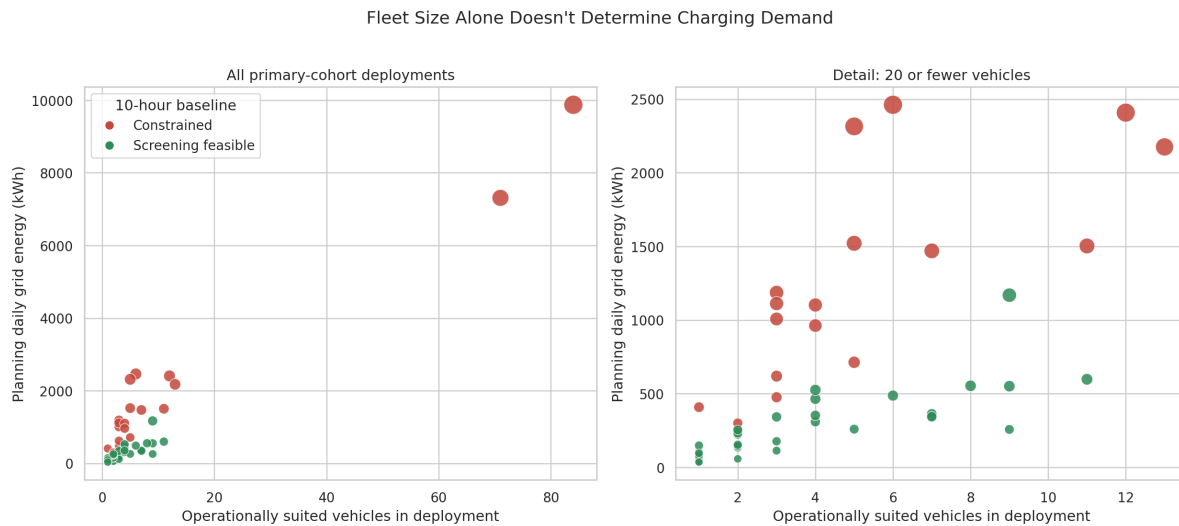
Figure 1
Planning Mileage Among Operationally Suited Vehicles



Note. Boxes summarize vehicle-level 90th-percentile daily mileage for the primary cohort. Outliers are suppressed in the display for readability, but they remain in the analysis.

The 53 primary-cohort deployments range from 36.47 to 9,873.38 kilowatt-hours of planning energy per day, with a median of 353.12. Figure 2 shows that larger fleets generally require more energy, but fleet size doesn't determine demand by itself. Deployments of similar size can differ because their vehicles travel different distances and use different class-level energy factors. The two largest deployments also stretch the scale, so the dense lower-left portion should be read as a screening distribution rather than a precise linear relationship.

Figure 2
Fleet Size Alone Does Not Determine Planning Energy



Note. Each point is a primary-cohort deployment. Color shows screening feasibility under the 10-hour baseline, and point size reflects planning energy.

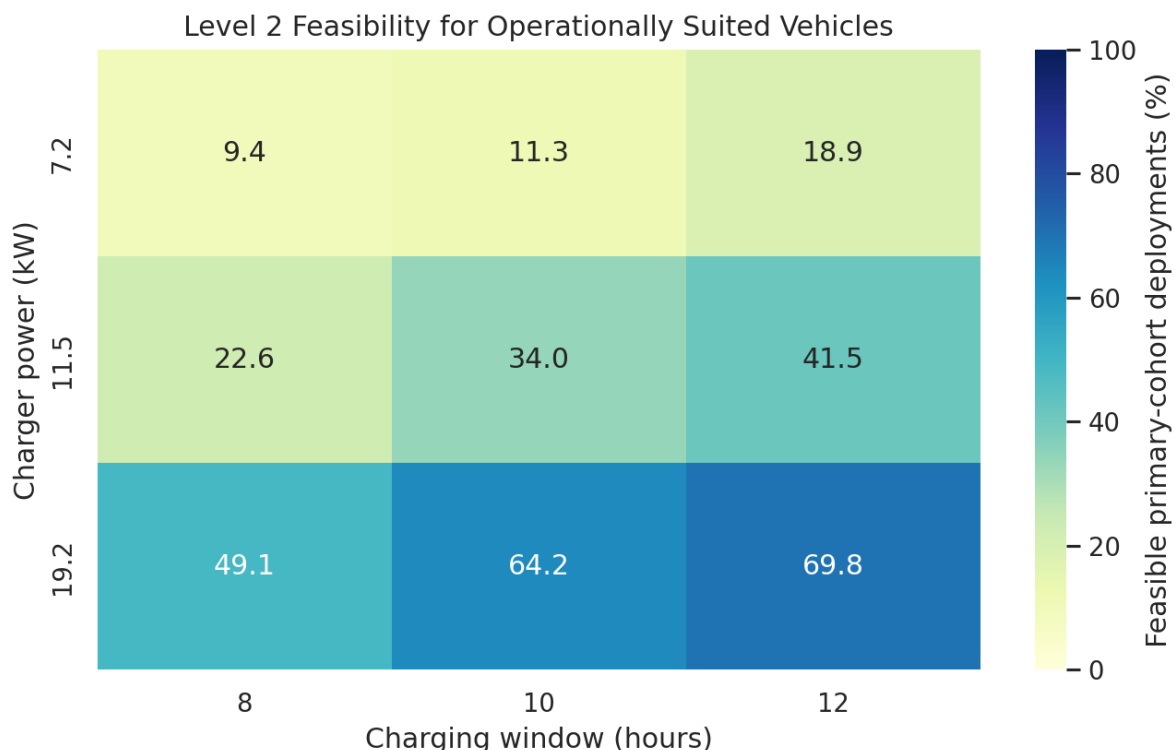
Reapplying the earlier operational screen changes the group of vehicles and deployments included in the infrastructure analysis, but it doesn't automatically increase the number of feasible deployments. Both the all-vehicle comparison and the primary cohort contain 34 screening-feasible deployments. The reported rate rises from 59.65% (34 of 57) to 64.15% (34 of 53) because four deployments drop out and other deployment-level changes offset one another. For example, Deployment 27 was feasible in the all-vehicle comparison but had no vehicles that passed the majority rule, while Deployment 50 changed from constrained to screening-feasible

as its selected vehicle count fell from three to two and its planning energy fell from 424.94 to 232.60 kilowatt-hours. The primary cohort's lower median energy shouldn't be treated as independent validation because mileage influences both the inherited suitability rules and the energy estimate.

Feasibility improves as either power or charging time increases. At 7.2 kilowatts, only 9.43% of deployments are screening-feasible with an eight-hour window and 18.87% with a 12-hour window. At 11.5 kilowatts, the rate rises from 22.64% to 41.51%. At 19.2 kilowatts, it reaches 49.06%, 64.15%, and 69.81% across the 8-, 10-, and 12-hour windows. Figure 3 shows that additional dwell time helps, but charger power has the larger effect within the tested range. Under the selected 10-hour baseline, 26 of the 34 feasible deployments require the 19.2-kilowatt option; five use 11.5 kilowatts and three use 7.2 kilowatts.

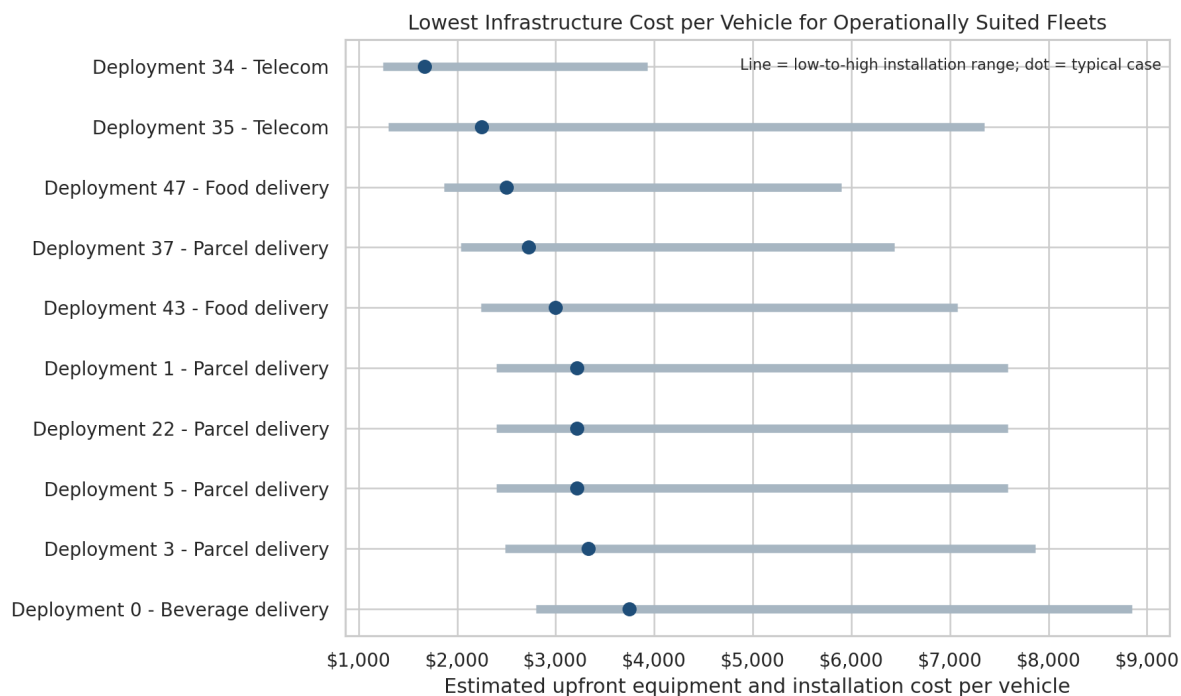
Figure 3

Screening Feasibility Improves With Charger Power and Time



Note. Cells show the percentage of the 53 primary-cohort deployments that pass the aggregate-energy, highest-demand-vehicle, and port-count rules.

For the 34 screening-feasible deployments, the median selected configuration uses two ports. The median typical upfront estimate is \$15,000, while the median low-to-high range is \$11,200 to \$35,400. On a per-vehicle basis, the median typical estimate is \$4,500, with median low and high estimates of \$2,800 and \$13,275. Figure 4 compares the ten lowest typical costs per vehicle, but the ranking isn't an electrification priority list. A low screening cost doesn't confirm available electrical capacity, operating value, vehicle replacement timing, or incentive eligibility.

Figure 4*Lowest Planning Cost per Vehicle Under the 10-Hour Baseline*

Note. Lines show low-to-high planning estimates and dots show the typical case. Estimates include charger equipment and assumed per-port installation only.

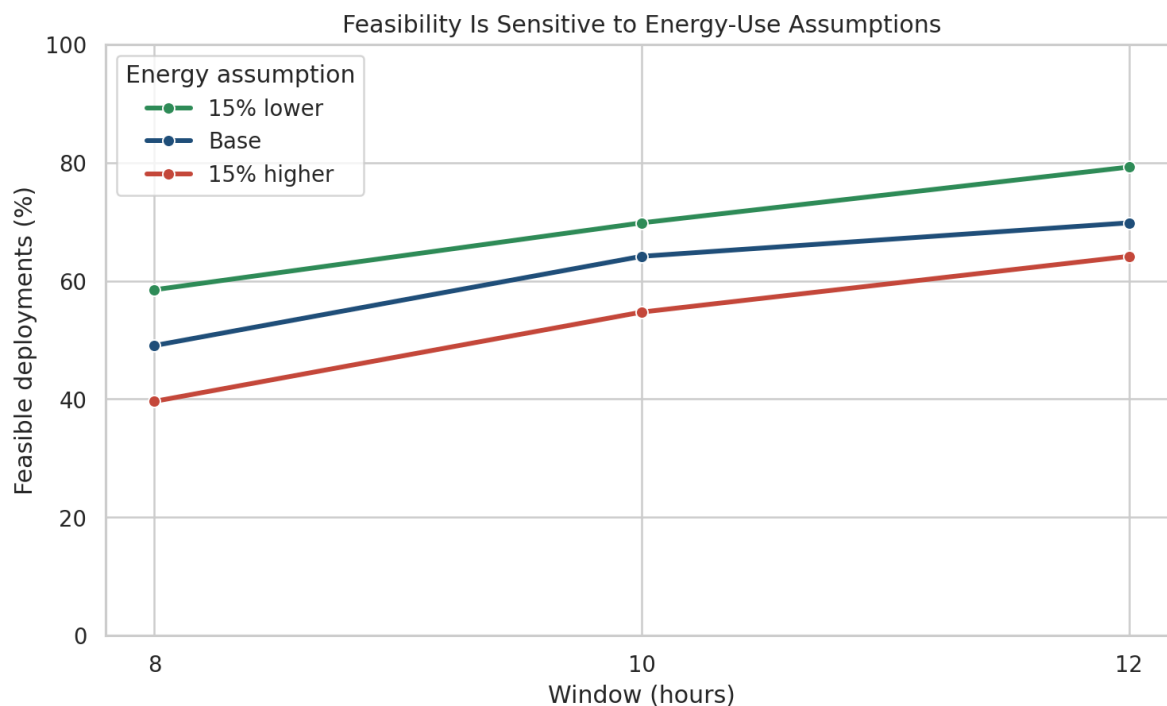
The model passed directional validation checks: outputs weren't missing, high installation cost exceeded typical cost, typical cost exceeded low cost, longer windows increased energy available from a port, and every selected baseline configuration met the stated screening rules. Because this isn't a predictive model, these checks and sensitivity analysis take the place of cross-validation or a confusion matrix. They confirm internal consistency, not real-world accuracy.

Figure 5 shows that the conclusion remains sensitive to vehicle energy use. With a 19.2-kilowatt charger and a 10-hour window, the screening-feasible share is 64.15% in the base case. It increases to 69.81% when energy use is 15% lower and falls to 54.72% when energy use is 15% higher. At eight hours, the range is 39.62% to 58.49%; at 12 hours, it is 64.15% to 79.25%.

A site near the feasibility boundary therefore needs vehicle-specific energy modeling before an investment decision.

Figure 5

Screening Feasibility Is Sensitive to Energy-Use Assumptions



Note. Lines show the percentage of primary-cohort deployments that pass the screening rules at 19.2 kilowatts when class-level energy factors are 15% lower, unchanged, or 15% higher.

Conclusion

This analysis demonstrates how a fleet manager can use vehicle operating data to begin planning for charging infrastructure. Within the Fleet DNA sample, Level 2 charging could support 34 of the 53 primary-cohort deployments under the 10-hour scenario, although most feasible deployments required 19.2-kilowatt chargers. The results also show why fleet size alone is not enough to estimate infrastructure need: deployments with similar vehicle counts can still have very different planning-energy requirements because mileage, vehicle class, and duty cycle matter.

The model gives fleet managers an early estimate of charging demand, required ports, charger power, and potential infrastructure costs. It can help identify which depots appear ready for a detailed site assessment and which may need longer charging windows, operational changes, managed charging, or a different Level 2 charging strategy. The results are intended to support early planning and should not replace utility review, electrical engineering, vendor quotes, or a detailed vehicle-to-port charging schedule.

Assumptions

The analysis assumes that the sampled days reasonably represent each vehicle's normal use and that the operational-suitability rules established in the earlier project provide a reasonable first-stage screen. Those rules aren't independently validated against actual electrification outcomes in this project. The analysis also assumes that a majority of highly suited days is enough to place a vehicle in the primary cohort, the 90th-percentile mileage is an appropriate planning value, the class-level energy factors are suitable for initial screening, charging is 90% efficient, 85% of the stated window is usable, and vehicles can share ports without the model constructing a detailed schedule. Cost assumptions are expressed in U.S. dollars and represent planning cases rather than quotes.

Limitations

Fleet DNA is a public composite dataset, not a census of every fleet or operating condition. Some vehicles have short histories, the vocation labels are broad, and the data don't include weather, payload, route elevation, vehicle model, state of charge, charger assignment, depot dwell time, electrical service capacity, or site layout. The operational rules and majority

threshold are analyst-defined and haven't been validated against actual electrification outcomes. The same mileage information influences both cohort selection and energy demand, so the lower demand in the selected cohort isn't independent confirmation of the screen. Finally, screening feasibility doesn't guarantee a workable charging schedule or a buildable site.

Challenges

The largest challenge was using the earlier vehicle-suitability results without making the current findings sound more certain than they are. To address this, the analysis keeps all usable operating days for the selected vehicles, compares them with the larger group of vehicles, and describes the results as an initial screening rather than a final decision. Missing site and cost information created another challenge. The model addresses this by showing low, typical, and high cost estimates and testing how changes in energy use affect the results. These steps don't eliminate the uncertainty, but they make the assumptions and limitations easier to understand.

Recommendations

Fleet managers should use the primary-cohort result as a project-screening tool. Deployments that pass the 10-hour Level 2 baseline, have reasonable planning costs, and can validate dwell time, parking, and operational assumptions should move to detailed site review. Constrained deployments should pause or redesign before moving forward. That may include testing longer charging windows, staggered charging, managed charging, or the 19.2-kilowatt Level 2 option before a deployment is removed from consideration. The model should also retain both the primary cohort and the all-vehicle comparison so decision makers can see when filtering changes a conclusion rather than assuming operational suitability and charging feasibility are the same decision.

Implementation Plan

Implementation can follow four stages. First, the fleet should replace the composite records with at least several weeks of its own telematics data and confirm the operational-suitability rules with operations staff. Second, the team should verify parking duration, shift overlap, and vehicle availability, then build an explicit vehicle-to-port schedule. Third, facilities and the serving utility should review electrical capacity, demand-management options, interconnection timing, code requirements, and a site-specific cost estimate. Finally, the fleet can pilot a small number of vehicles, compare measured energy use with the planning factors, and update the model before expanding. A project should advance only when operational, charging, financial, and site evidence agree.

Future Uses and Additional Applications

Future work can connect this screening model with the earlier operational analysis in a single decision pipeline and add vehicle replacement timing, fuel and maintenance savings, incentives, emissions, and utility rates. A scheduling optimizer could test whether individual vehicles can share chargers across arrival and departure times. Probabilistic simulation would become appropriate once measured distributions are available for energy use, dwell time, temperature, and installation cost. The same framework could also support phased infrastructure planning by identifying which deployments remain feasible as fleet size grows.

Ethical Assessment

Identify

The main ethical concerns are data validity, representativeness, false precision, and misuse. Fleet DNA is anonymized public data and doesn't expose driver identities, so privacy

risk is low. However, the sample may not represent every region, climate, vocation, or operating practice. The analysis also transforms observed mileage into estimated energy and cost, which can look more certain than the source data justify. A decision maker could misuse the ranking to approve or reject projects without a site review.

Assess

Model uncertainty is highly likely because several important variables are absent; its impact could be high if the results are treated as an engineering design or budget.

Representativeness concerns are moderately likely and could affect fleets whose operations differ from the composite sample. Privacy harm is unlikely because the source is already anonymized and the analysis doesn't attempt re-identification. Legal and regulatory risk is also high if a screening result is mistaken for compliance with electrical codes, accessibility requirements, permitting, utility interconnection, or workplace rules.

Mitigate

The paper labels every modeled quantity as a planning estimate, documents all filters and assumptions, reports sensitivity rather than one fixed answer, and preserves an all-vehicle comparison. No records were altered to force a preferred result, and exclusions are counted explicitly. A future implementation should show source dates, assumption ranges, and a prominent screening-only warning; restrict decisions until operations, facilities, utility, and engineering reviewers confirm the result; and test performance across vocations and regions before broader use. The Fleet DNA source and supporting federal guidance were obtained from public, credible sources and used for the educational purpose described here.

Code and Reproducibility

The accompanying notebook, DSC680_Milestone2_Morrow_D.ipynb, contains the complete reproducible workflow, including data preparation, cohort selection, charging scenarios, cost estimates, validation, sensitivity analysis, and figure generation. The notebook is submitted with the final white paper so the reported results can be reproduced from the analysis code.

Appendix A: Supporting Documentation

Key Data Fields

Field	Meaning	Use in Analysis
vid	Anonymous vehicle identifier	Vehicle-level grouping
did	Anonymous deployment identifier	Deployment-level grouping
day_id	Observed vehicle day	Duplicate checks and day counts
class_id	Vehicle weight class	Energy-use factor
voc_id	Anonymous vocation code	Readable vocation grouping
distance_total	Daily miles traveled	Inherited suitability screen and planning mileage
driving_time	Daily driving duration	Inherited suitability rule
average_speed	Average daily speed	Inherited suitability rule
percent_zero	Share of time at zero speed	Inherited route-intensity rule
stops_per_mile	Stops divided by daily miles	Inherited route-intensity rule
trips / stops	Daily trip and stop counts	Inherited route-intensity rule

Model Assumptions and Decision Rules

Item	Value or Rule
Minimum history	At least three recorded days
Earlier-project day rule	At least four of five inherited suitability rules
Current-project vehicle rule	More than 50% of recorded days highly suited
Planning mileage	Vehicle-level 90th percentile across all usable days

Item	Value or Rule
Class energy factors	0.55, 0.70, 0.90, 1.10, 1.30, 1.70, and 2.10 kWh/mile for Classes 2–8
Charging setup	90% efficiency; 85% of an 8-, 10-, or 12-hour window; 7.2, 11.5, or 19.2 kW
Screening feasible	Aggregate energy, highest-demand vehicle, and port-count rules all pass
Baseline selection	Lowest typical cost at 10 hours; ties favor less connected power
Cost inputs	Equipment: \$2,000, \$3,500, or \$5,000 per port; installation: \$600 low, \$2,500 typical, or \$12,700 high per port

Core Calculations

Grid energy per vehicle = planning mileage × class kWh/mile ÷ charging efficiency.

Effective energy per port = charger power × charging window × usable-window share.

Required ports = deployment planning energy ÷ effective energy per port, rounded up.

Typical upfront cost = required ports × (equipment cost per port + typical installation cost per port).

Appendix B: Audience Questions and Answers

1. How were the operational-suitability results from the earlier project incorporated into this charging infrastructure analysis?

Answer: The earlier suitability rules were reapplied to the Fleet DNA vehicle-day records to identify a primary cohort for charging analysis. A vehicle entered the primary cohort when more than half of its usable days met the high-suitability definition. After selection, the charging requirement used all usable days for that vehicle, not only the favorable days, so harder operating days were still represented.

2. Why did the model use the 90th percentile instead of the average or maximum daily mileage?

Answer: The average can understate planning need because it smooths over demanding days, while the maximum can size the infrastructure around a single extreme day. The 90th percentile covers most observed operating days without making every depot plan for the highest one-day value in the data.

3. Where did the class-level kilowatt-hour-per-mile factors come from, and how should they be validated?

Answer: The factors are AFLEET-informed planning assumptions by vehicle class. They are useful for a first-pass conversion from mileage to energy, but they should be validated with measured energy use from the fleet's actual vehicles because payload, weather, terrain, auxiliary loads, and driver behavior can change consumption.

4. Why was 10 hours used as the baseline if Fleet DNA does not contain complete dwell-time data?

Answer: Fleet DNA does not provide complete arrival and departure schedules, so the analysis tested 8-, 10-, and 12-hour windows. The 10-hour case is the midpoint baseline for comparison, not a claim that every deployment actually has 10 available hours.

5. What does screening-feasible mean, and how is it different from an approved charging design?

Answer: Screening-feasible means the scenario passed three planning checks: aggregate energy, highest-demand vehicle energy, and port count. It does not prove that the site is buildable, that a charging schedule works, or that electrical service, permitting, utility requirements, and layout constraints have been resolved.

6. Could a deployment pass the model but fail once individual arrival and departure times are scheduled?

Answer: Yes. The model does not build a vehicle-to-port charging schedule. A deployment could pass the aggregate planning checks and still fail once exact return times, shift overlap, parking assignments, charger sharing, and plug-in behavior are modeled.

7. How would weather, payload, terrain, and auxiliary loads change the energy estimate?

Answer: Those factors could increase or decrease actual energy use relative to the class-level assumptions. Cold weather, heavy payloads, hilly terrain, HVAC loads, and auxiliary equipment can raise demand, which is why the sensitivity analysis tested the effect of higher and lower energy-use assumptions.

8. Which site costs are missing from the low, typical, and high estimates?

Answer: The estimates include charger equipment and assumed per-port installation ranges. They do not include a site-specific electrical service upgrade, transformer work, unusual trenching, parking reconstruction, permitting, networking, demand-management equipment, utility make-ready work, or final vendor pricing.

9. Why did the primary cohort have the same number of feasible deployments as the all-vehicle comparison?

Answer: The primary cohort changed which vehicles and deployments were included, but it did not automatically increase the number of feasible deployments. Some deployments dropped out when no vehicles met the majority rule, while other deployments became less demanding after selected vehicle counts and planning energy changed. The feasible count stayed at 34, while the denominator changed from 57 to 53.

10. What additional information is needed before a fleet purchases vehicles or charging equipment?

Answer: The fleet should confirm actual return times, parking layout, charging-window assumptions, charger assignments, electrical service capacity, utility requirements, permits, vendor quotes, and vehicle-specific energy use. The model supports early screening, but purchasing decisions should wait until operations, facilities, utility, engineering, and financial evidence agree.

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